

Supplementary Materials: LLM-Assisted Extraction of QSP Calibration Targets

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Contents

1	S1: Complete SimplifiedIsolatedTarget Example	2
1.1	Overview	2
1.2	Complete YAML Specification	2
1.3	Schema Structure Summary	5
2	S2: Supported Model Types	5
3	S3: Validation Checks	6
4	S4: Generated Julia Code Structure	6

1 S1: Complete SimplifiedIsolatedTarget Example

This section presents a complete calibration target for activated pancreatic stellate cell (aPSC) proliferation, demonstrating all schema components.

1.1 Overview

This target constrains the proliferation rate parameter `k_apsc_prolif` using data from Schneider et al. (2001), who measured PSC proliferation in response to PDGF stimulation via BrdU incorporation assay.

1.2 Complete YAML Specification

```
1 # Schema: SimplifiedIsolatedTarget
2 # Target: Activated PSC proliferation rate
3
4 target_idpsc_proliferation_PDAC_deriv001
5
6 # =====
7 # INPUTS: Extracted values with full provenance
8 # =====
9 inputs:
10   -name: pdgf_fold_mean
11     value: 4.37
12     units: dimensionless
13     input_type: direct_measurement
14     role: target
15     source_ref: Schneider2001_PSC_PDGF
16     source_location: Abstract
17     value_snippet: "Cell proliferation (4.37 +/- 0.49- and 2.96 +/-
18                   0.39-fold of control).\"
19
20   -name: pdgf_fold_sd
21     value: 0.49
22     units: dimensionless
23     input_type: direct_measurement
24     role: auxiliary
25     source_ref: Schneider2001_PSC_PDGF
26     source_location: Abstract
27     value_snippet: "Cell proliferation (4.37 +/- 0.49- and 2.96 +/-
28                   0.39-fold of control).\"
29
30 # =====
31 # CALIBRATION: Everything needed for inference
32 # =====
33 calibration:
34   parameters:
35     -name: k_apsc_prolif
36       units: 1/day
37       prior:
38         distribution: lognormal
39         mu: 0.0 # log(1.0) -expect ~1/day for 4x fold-change in 1 day
40         sigma: 1.0
41         rationale: |
42           Wide log-normal prior centered at 1/day. A 4-fold increase in
43           1 day implies  $k = \ln(4) = 1.4/\text{day}$ , so prior median of 1/day
44           is reasonable.
45
46   state_variables:
47     -name: aPSC_rel
48       units: dimensionless
49       initial_condition:
50         value: 1.0
```

```

51     rationale: |
52         PSC proliferation data are reported as fold of control. We
53         normalize the initial aPSC population to 1.0 (100% control)
54         to match this convention.
55
56 model:
57     type: exponential_growth
58     rate_constant: k_apsc_prolif
59     data_rationale: |
60         BrdU incorporation assay measures proliferation as fold-change
61         over ~24 hours under constant PDGF stimulus. With a single high
62         PDGF dose and short duration, exponential growth ( $dN/dt = k \cdot N$ )
63         fits the data structure.
64     submodel_rationale: |
65         The full QSPI0_PDAC model has aPSC proliferation via:
66          $k\_apsc\_prolif * aPSC * (PDGF / (PDGF\_50\_prolif + PDGF))$ .
67         At 50 ng/mL PDGF  $\gg$  PDGF_50_prolif, the saturation term  $\sim 1$ ,
68         reducing to exponential growth with rate k_apsc_prolif.
69
70 independent_variable:
71     name: time
72     units: day
73     span: [0.0, 1.0]
74     rationale: |
75         Proliferation assay duration assumed to be ~24 hours based on
76         typical cytokine incubation protocols for rat PSCs.
77
78 measurements:
79     -name: proliferation_fold_change
80       observable:
81         type: identity
82         state_variables: [aPSC_rel]
83         rationale: |
84             The state variable aPSC_rel is the fold-change in aPSC number
85             relative to baseline (control = 1.0), which directly matches
86             the reported proliferation fold-change.
87         units: dimensionless
88         uses_inputs:
89             -pdgf_fold_mean
90             -pdgf_fold_sd
91         evaluation_points: [1.0]
92         sample_size: 3
93         sample_size_rationale: |
94             Sample size not explicitly reported; assumed n=3 independent
95             experiments, typical for in vitro PSC BrdU proliferation assays.
96         distribution_code: |
97             def derive_distribution(inputs, ureg):
98                 import numpy as np
99
100                 rng = np.random.default_rng(42)
101
102                 mean = inputs['pdgf_fold_mean'].magnitude
103                 sd = inputs['pdgf_fold_sd'].magnitude
104                 n = 10000
105
106                 # Use lognormal for positive fold-changes
107                 mu_log = np.log(mean**2 / np.sqrt(mean**2 + sd**2))
108                 sigma_log = np.sqrt(np.log(1 + sd**2 / mean**2))
109
110                 samples = rng.lognormal(mu_log, sigma_log, n)
111
112                 median_val = np.median(samples)
113                 ci_lower = np.percentile(samples, 2.5)
114                 ci_upper = np.percentile(samples, 97.5)
115
116                 return {
117                     'median': [median_val],
118                     'ci95': [[ci_lower, ci_upper]],

```

```

119         'units': 'dimensionless'
120     }
121     likelihood:
122         distribution: lognormal
123         rationale: |
124             Fold-changes are positive and multiplicative, making
125             lognormal appropriate.
126
127     identifiability_notes: |
128         With a single timepoint (~24h) and normalized initial condition
129         (aPSC_rel(0)=1), only k_apsc_prolif is estimated. If substantial
130         PSC death occurred over the assay interval, the inferred rate
131         would represent net growth rather than pure proliferation.
132
133     # =====
134     # EXPERIMENTAL CONTEXT
135     # =====
136     experimental_context:
137         species: rat
138         system: in_vitro_primary_cells
139         indication: PDAC
140         cell_types:
141             -name: pancreatic stellate cell
142               phenotype: activated (alpha-SMA positive)
143               isolation_method: Isolated from normal Wistar rat pancreas,
144                             activated by culture on plastic
145         culture_conditions:
146             culture_type: 2d_monolayer
147
148     # =====
149     # STUDY INTERPRETATION
150     # =====
151     study_interpretation: |
152         Schneider et al. (2001) measured proliferation of cultured rat
153         pancreatic stellate cells (PSCs) in response to growth factors using
154         BrdU incorporation. Addition of 50 ng/mL PDGF increased PSC
155         proliferation to 4.37 +/- 0.49-fold of control, indicating that PDGF
156         acts as a strong mitogen for activated PSCs. We interpret the
157         50 ng/mL PDGF condition as approximating a near-maximal PDGF stimulus
158         and use the reported fold-change over ~24 hours as a quantitative
159         constraint on the maximal PDGF-driven proliferation rate parameter
160         k_apsc_prolif.
161
162     key_assumptions:
163         -|
164             50 ng/mL PDGF corresponds to a near-saturating concentration for
165             PSC proliferation, so  $PDGF/(PDGF_{50\_prolif} + PDGF) \sim 1$ .
166         -|
167             The proliferation assay duration is approximately 24 hours,
168             consistent with typical cytokine incubation protocols.
169         -|
170             BrdU incorporation fold-change is proportional to fold-change in
171             PSC cell number (negligible death over 24 hours).
172         -|
173             Rat activated PSC proliferation kinetics in vitro provide an upper
174             bound for human tumor aPSC proliferation in vivo.
175
176     key_study_limitations:
177         -|
178             Proliferation was measured via BrdU incorporation rather than
179             direct cell counts.
180         -|
181             Single PDGF concentration and single assay interval constrain
182             only the maximal rate.
183         -|
184             Data from rat PSCs in vitro, not human PDAC-associated stellate cells.
185         -|
186             Background serum proliferative signals not explicitly modeled.

```

```

187
188 # =====
189 # DATA SOURCES
190 # =====
191 primary_data_source:
192   doi: "10.1152/ajpcell.2001.281.2.C532"
193   title: "Identification of mediators stimulating proliferation and
194         matrix synthesis of rat pancreatic stellate cells"
195   authors: [Schneider, Schmid, Liptay, Schmid, Pohl]
196   year: 2001
197   source_tag: Schneider2001_PSC_PDGF
198
199 secondary_data_sources:
200   -doi: "10.1136/gut.44.4.534"
201     title: "Pancreatic stellate cells are activated by proinflammatory
202           cytokinesimplications for pancreatic fibrogenesis"
203     authors[Apte, Haber, Darby, Rodgers, McCaughan, Korsten, Pirola, Wilson]
204     year1999
205     source_tagApte1999_PSC_Cytokines
206     contribution|
207       Provides context on PSC activation and cytokine response protocols.

```

1.3 Schema Structure Summary

The SimplifiedIsolatedTarget schema has the following top-level structure:

- **target_id**: Unique identifier for the calibration target
- **inputs**: List of values extracted from literature with provenance
- **calibration**: Everything needed for inference code generation
 - **parameters**: Parameters to estimate with priors
 - **state_variables**: ODE state variables with initial conditions
 - **model**: Model type specification (built-in or custom)
 - **independent_variable**: Time or other independent variable
 - **measurements**: Observables with likelihoods
 - **identifiability_notes**: Discussion of parameter identifiability
- **experimental_context**: Species, system, cell types, culture conditions
- **study_interpretation**: Scientific interpretation of the data
- **key_assumptions**: Assumptions made in using this data
- **key_study_limitations**: Known limitations of the source study
- **primary_data_source**: Primary literature reference with DOI
- **secondary_data_sources**: Additional references

2 S2: Supported Model Types

The schema supports the following built-in model types:

Table 1: Built-in model types with their ODE formulations.

Model Type	ODE	Parameters
exponential_growth	$dy/dt = k \cdot y$	rate_constant
first_order_decay	$dy/dt = -k \cdot y$	rate_constant
logistic	$dy/dt = k \cdot y \cdot (1 - y/K)$	rate_constant, carrying_capacity
saturation	$dy/dt = k \cdot (1 - y)$	rate_constant
two_state	$dA/dt = -k \cdot A, dB/dt = k \cdot A$	forward_rate
michaelis_menten	$dy/dt = -V_{max} \cdot y/(K_m + y)$	vmax, km
direct_conversion	(analytical formula)	formula
direct_fit	(curve fitting)	curve
custom	(user-provided)	code, code_julia

3 S3: Validation Checks

The validation framework includes the following automated checks:

1. **Input reference validation:** All `uses_inputs` references match defined input names
2. **Source reference validation:** All `source_ref` values match defined source tags
3. **Parameter reference validation:** Model parameter roles reference defined parameters
4. **State variable reference validation:** Observable `state_variables` match defined state variables
5. **DOI resolution:** All DOIs resolve via CrossRef API
6. **Value-in-snippet validation:** Extracted values appear verbatim in quoted snippets
7. **Unit validation:** All unit strings parse with Pint
8. **Code syntax validation:** Custom code passes Python AST parsing
9. **Span ordering validation:** Time spans have start < end
10. **ODE model requirements:** ODE models have required state variables and spans

4 S4: Generated Julia Code Structure

The translator generates Julia scripts with the following structure:

```

1 using DifferentialEquations
2 using Turing
3 using Distributions
4
5 # Observed data (from inputs layer)
6 const obs_proliferation_median = 4.37
7 const obs_proliferation_sigma = 0.25 # derived from CI95
8
9 # ODE function
10 function ode_proliferation!(du, u, p, t)
11     k = p[1]
12     du[1] = k * u[1] # exponential growth

```

```

13 end
14
15 # Simulate function
16 function simulate_proliferation(k; t_span=(0.0, 1.0), u0=[1.0])
17     prob = ODEProblem(ode_proliferation!, u0, t_span, [k])
18     sol = solve(prob, Tsit5())
19     return sol[end][1] # final value
20 end
21
22 # Turing model
23 @model function proliferation_model()
24     # Prior
25     k_apsc_prolif ~ LogNormal(0.0, 1.0)
26
27     # Simulate
28     pred = simulate_proliferation(k_apsc_prolif)
29
30     # Likelihood
31     obs_proliferation_median ~ LogNormal(log(pred), obs_proliferation_sigma)
32 end
33
34 # Run inference
35 model = proliferation_model()
36 chain = sample(model, NUTS(), MCMCThreads(), 1000, 4)

```

For joint inference over multiple targets, the translator identifies shared parameters by name and generates a single model with one prior per unique parameter.